Ops let's write title later

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Section 1: Introduction

Background of the Data

This data set contains weekly retail prices for six fuel types: Regular Unleaded Gasoline, Mid-Grade Gasoline, Premium Gasoline, Diesel, Auto Propane, and Compressed Natural Gas across 10 markets in Ontario. Prices are expressed in Canadian cents per litre, with compressed natural gas measured in cents per gasoline-equivalent litre. The data set spans from January 3, 1990 to March 16, 2026 and is updated weekly with the latest retail fuel price data published by the Government of Ontario.

Research Question

How does the price differ between cities in Ontario?

Section 2: Data Description

The data set contains 11,252 weeks of data from 1990 to March 16, 2026, covering six fuel types. Each row represents a weekly average of fuel prices, so daily fluctuations are not captured. It also includes three regional average columns (Ontario, Northern Ontario, Southern Ontario), which do not correspond to the simple average of the included cities, indicating they were sourced externally.

Section 3: Data Preparation/Cleaning

Prior to analysis, it is important to highlight certain features of the raw data set that affect how it should be handled:

1. Bilingual Structure of the Data Set

The data set contains both French and English labels, sometimes separated by slashes or presented in different columns. To maintain consistency, variables are standardized into a single language format before analysis.

We first rename the columns we keep.

Table 1: Old Column Names vs. New Column Names

Before	After
Date	Date
Ottawa	Ottawa

Before	After
Toronto.West.Ouest	Toronto_West
Toronto.East.Est	Toronto_East
Windsor	Windsor
London	London
Peterborough	Peterborough
St..Catharine.s	St_Catharines
Sudbury	Sudbury
Sault.Saint.Marie	Sault_Saint_Marie
Thunder.Bay	Thunder_Bay
North.Bay	North_Bay
Timmins	Timmins
Kenora	Kenora
Parry.Sound	Parry_Sound
Ontario.Average.Moyenne.provinciale	Ontario_Average
Southern.Average.Moyenne.du.sud.de.l.Ontario	Southern_Ontario_Average
Northern.Average.Moyenne.du.nord.de.l.Ontario	Northern_Ontario_Average
Fuel.Type	Fuel_Type
Type.de.carburant	Type.de.carburant

We also remove the column "Type.de.carburant" because it was the same as the information in the Fuel_Type column. More specifically, it contained the fuel type names in French.

2. Handling of Zero Values

Values recorded as 0 in the city-level columns are interpreted as missing data, since these 0s do not represent actual prices. To prevent these zeros from affecting computations, all 0 values are converted to NA prior to analysis.

3. Conversion of the Date Column

The Week column, originally recorded as a character string, was converted to a Date type to ensure proper chronological ordering and accurate calculations. This allows for correct filtering, aggregation, and plotting over time, and ensures that weekly comparisons between cities are reliable.

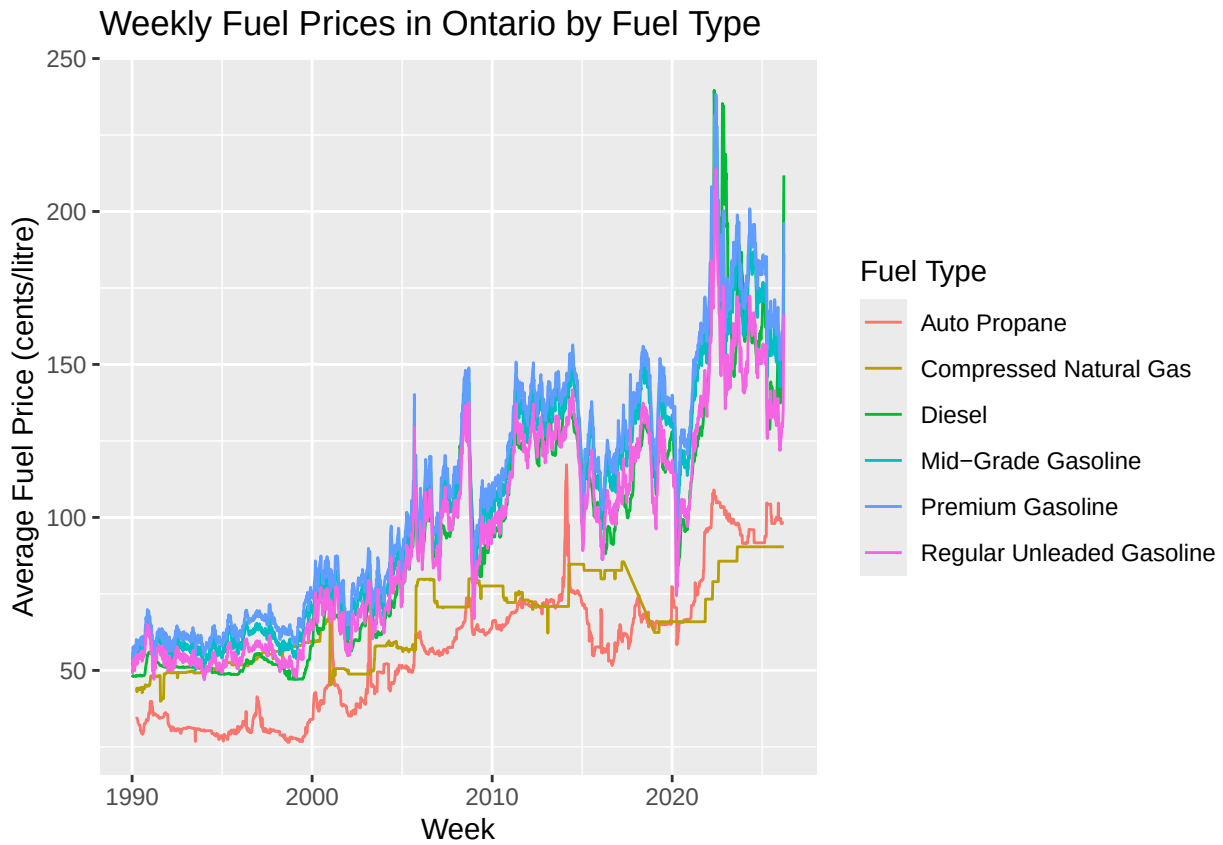
4. Categorization of the Data Set

As mentioned above, the data set includes six fuel types. To enable clearer and more accurate comparisons between cities, the data will be analyzed separately for each fuel type.

Section 4: Exploratory Data Analysis

Visualizations

We want



Section 5: Statistical Inference

To answer our research question on how prices differ across cities, we will compare the Northern Ontario Average price to the Southern Ontario Average price. To ensure our analysis provides a comprehensive view of the transportation sector, we will focus our statistical inference on four distinct fuel categories: Regular Unleaded Gasoline (representing the majority of consumer vehicles), Diesel (representing commercial transit), as well as Auto Propane (representing alternative fuels). Additionally, we have restricted our analysis to recent data (2021–2026) to ensure our findings reflect the current economic landscape and provide the most relevant insights. Because prices are recorded on the exact same weeks, we will analyze the paired differences (Northern Average - Southern Average).

5.1 Confidence Interval

We estimate difference in gas prices (Northern Average - Southern Average) across the available weeks between 2021-2026 for Regular Unleaded Gasoline, Diesel, .

Table 2: Average Price Differences: Northern vs. Southern Ontario

Fuel Type	Mean Difference (cents/litre)	95% Confidence Interval
Regular Unleaded Gasoline	2.57	(1.85, 3.29)
Diesel	4.30	(3.82, 4.77)
Auto Propane	19.58	(18.58, 20.59)

Interpretation:
Based on data from the last five years, we are 95% confident that the true average difference in fuel prices between Northern and Southern Ontario falls within the following intervals (in cents per litre):

- **Regular Gas:** 1.85 to 3.29
- **Diesel:** 3.82 to 4.77
- **Auto Propane:** 18.58 to 20.59

5.2 Hypothesis Testing

To rigorously answer our research question regarding regional price differences, we conducted a paired two-sample t-test for each of our three fuel types. We set our significance level at $\alpha = 0.05$.

Hypotheses: Let μ_d represent the true mean difference in weekly fuel prices between Northern and Southern Ontario ($\mu_{North} - \mu_{South}$).

- **Null Hypothesis (H_0):** $\mu_d = 0$. There is no true mean difference in fuel prices between Northern and Southern Ontario. (Any observed difference is due to random market fluctuations).
- **Alternative Hypothesis (H_1):** $\mu_d \neq 0$. There is a true, statistically significant mean difference in fuel prices between Northern and Southern Ontario.

Results:

Table 3: Paired T-Test Results: Regional Price Differences

Fuel Type	p-value
Regular Unleaded Gasoline	1.8073e-11
Diesel	$<$ 2.22e-16
Auto Propane	$<$ 2.22e-16

Conclusion:
As shown in the table above, the calculated p-values for all four fuel types—Regular Unleaded Gasoline, Diesel, Auto Propane, and Compressed Natural Gas—are strictly less than our significance level of 0.05. Because of this, **the price differences for every fuel category are statistically significant,** and **we confidently reject the null hypothesis (H_0) in all cases.**

Section 6: Bootstrapping

While the paired t-tests in Section 5 provided strong evidence of a regional price divide, parametric tests rely on the assumption that the sample means are normally distributed. Real-world commodities like fuel are subject to sudden supply shocks and extreme price volatility, which can introduce outlines into the data.

To ensure our findings are robust and not skewed by these potential outliers, we utilize a non-parametric approach: **Bootstrapping**. By resampling our modern (2021-2026) paired regional price differences 1,000 times with replacement, we can generate an empirical distribution of the mean difference without relying on assumptions of normality.

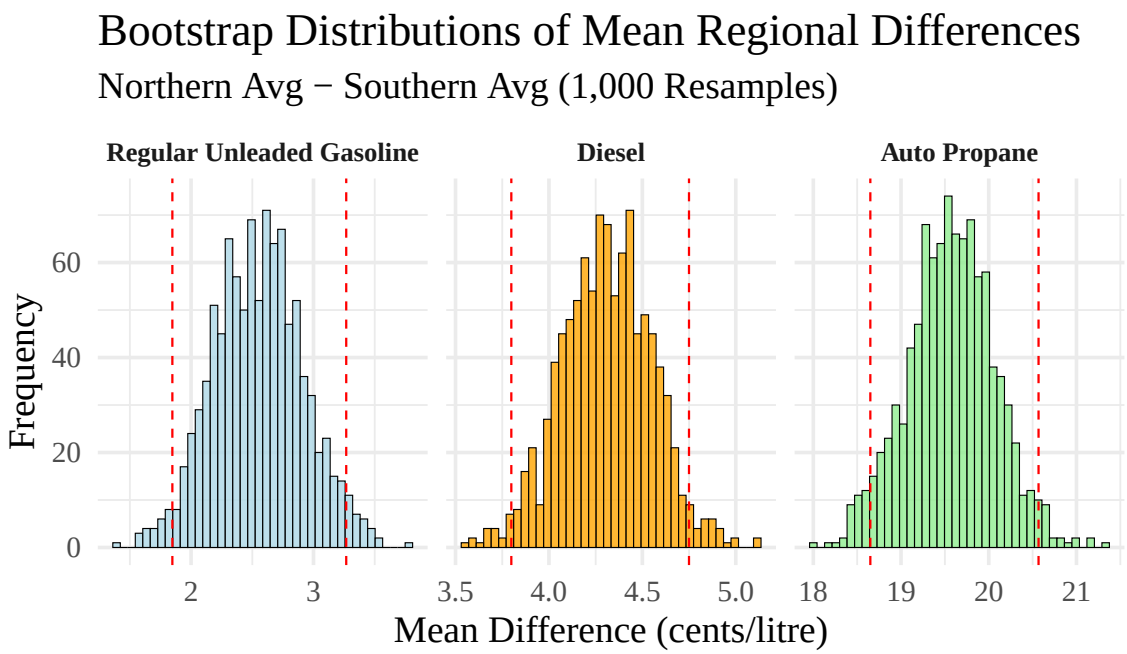
6.1 Bootstrap Confidence Intervals vs. Calssical Intervals

Table 4: Comparison of Classical and Bootstrapped 95% Confidence Intervals

Fuel Type	Classical 95% CI (T-Test)	Bootstrap 95% CI
Regular Unleaded Gasoline	(1.85, 3.29)	(1.85, 3.27)
Diesel	(3.82, 4.77)	(3.8, 4.75)
Auto Propane	(18.58, 20.59)	(18.65, 20.57)

6.2 Visualization of Bootstarpped Means

To better conceptualize this process, we can visualize the 1,000 resampled mean differences for Regular Unleaded Gasoline. The red dashed lines represent the boundaries of our 95% Bootstrap Confidence Interval.



6.3 Interpretation

As demonstrated in the comparison table, our non-parametric Bootstrap confidence intervals are nearly identical to the classical parametric intervals calculated in Section 5. Furthermore, the visualized distributions of the resampled means for all three fuel types exhibit clear, bell-shaped curves that sit entirely above zero.

Section 7: Regression Analysis

In order to better understand the fuel price difference between the cities, we fit a regression model with time and city as predictors. From the weekly fuel price plot in Section 4, we can clearly see the correlation between time and fuel price is non-linear. Thus, we fit a polynomial regression model.

To prevent overfitting, we build the full model of maximum degree 5, and use cross-validation to determine the best degree model. To make sure the model perform well throughout this time period, we use 5-fold cross-validation.

Table 5: Cross-Validation Error (Regular Gas)

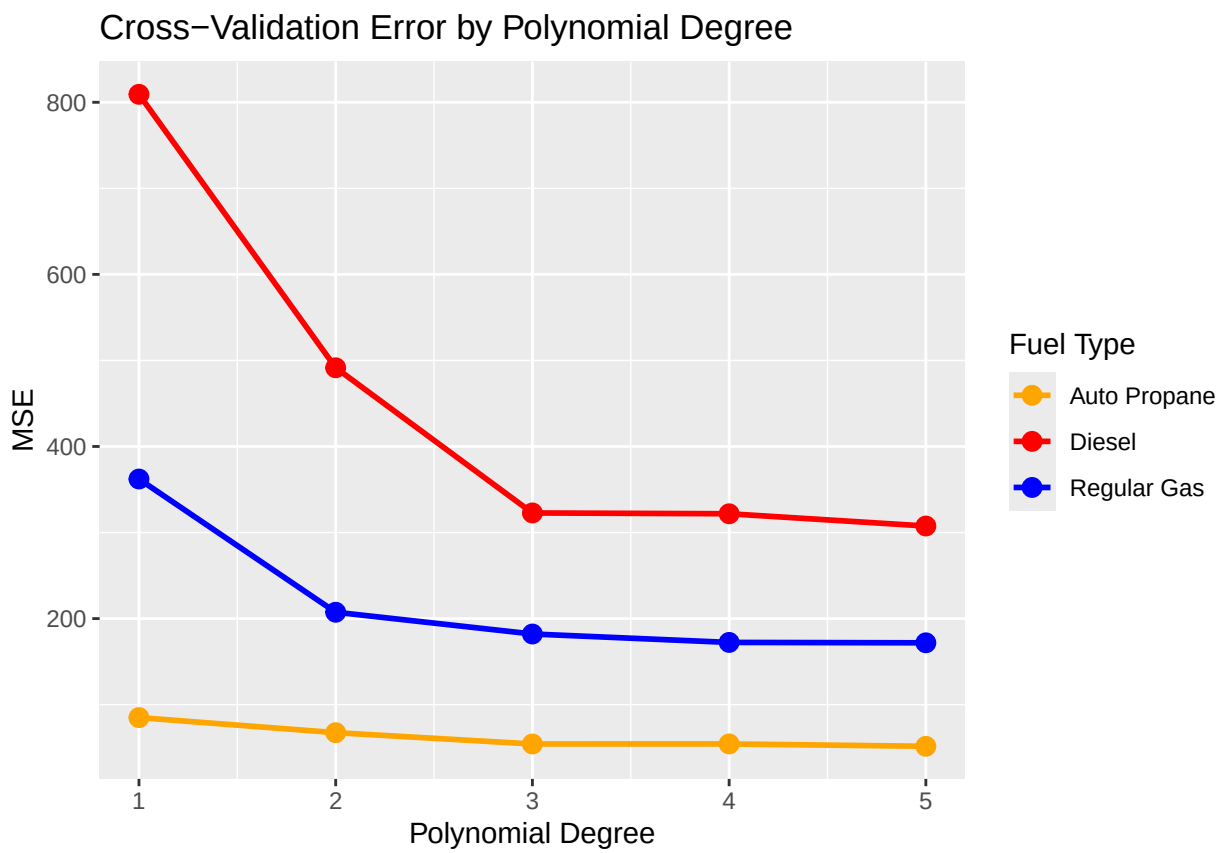
Polynomial Degree	Cross-Validated MSE
Degree 1	362.1822
Degree 2	207.3702
Degree 3	182.0726
Degree 4	172.2935
Degree 5	171.8376

Table 6: Cross-Validation Error (Diesel)

Polynomial Degree	Cross-Validated MSE
Degree 1	809.2580
Degree 2	491.4847
Degree 3	322.7642
Degree 4	321.8114
Degree 5	307.5842

Table 7: Cross-Validation Error (Auto Propane)

Polynomial Degree	Cross-Validated MSE
Degree 1	84.94025
Degree 2	67.41737
Degree 3	54.26658
Degree 4	54.32162
Degree 5	51.56948



We can see for all fuel types, from degree 1 to degree 3, MSE dropped significantly because of the increase in model complexity. However, after degree 3, the extra accuracy is marginal. Since we prefer simpler model at similar accuracy, we choose degree 3 model as the final model.

```
##
## Call:
## lm(formula = Price ~ City + poly(Time_Index, 3), data = regular_gas_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -36.148  -8.387  -1.666   6.863  60.144
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)            156.3092     0.8155 191.674  < 2e-16 ***
## CityLondon              -7.1397     1.1533  -6.191 6.63e-10 ***
## CityNorth_Bay           -6.0956     1.1533  -5.285 1.32e-07 ***
## CityOttawa              -7.9864     1.1533  -6.925 5.10e-12 ***
## CityParry_Sound         -7.8309     1.1533  -6.790 1.30e-11 ***
## CityPeterborough       -14.3669     1.1533 -12.457  < 2e-16 ***
## CitySault_Saint_Marie   -6.9379     1.1533  -6.016 1.96e-09 ***
## CitySt_Catharines       -9.3978     1.1533  -8.149 4.95e-16 ***
## CitySudbury             -6.8945     1.1533  -5.978 2.47e-09 ***
## CityThunder_Bay         -3.2702     1.1533  -2.836   0.0046 **
## CityTimmins              0.8665     1.1533   0.751   0.4525
## CityToronto_East        -7.2460     1.1533  -6.283 3.70e-10 ***
```

```
## CityToronto_West        -6.7298     1.1533  -5.835 5.82e-09 ***
## CityWindsor             -8.2640     1.1533  -7.166 9.26e-13 ***
## poly(Time_Index, 3)1  -142.2913    13.4495 -10.580  < 2e-16 ***
## poly(Time_Index, 3)2  -763.4847    13.4495 -56.767  < 2e-16 ***
## poly(Time_Index, 3)3   310.1864    13.4495  23.063  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.45 on 3791 degrees of freedom
## Multiple R-squared:  0.5223, Adjusted R-squared:  0.5203
## F-statistic: 259.1 on 16 and 3791 DF,  p-value: < 2.2e-16
```

```
##
## Call:
## lm(formula = Price ~ City + poly(Time_Index, 3), data = diesel_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -38.995 -11.422  -1.456   9.439  67.411
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)             167.1912     1.0852 154.067  < 2e-16 ***
## CityLondon               -8.4243     1.5347  -5.489 4.30e-08 ***
## CityNorth_Bay            -5.1445     1.5347  -3.352 0.000810 ***
## CityOttawa               -5.4596     1.5347  -3.557 0.000379 ***
## CityParry_Sound          -7.4728     1.5347  -4.869 1.17e-06 ***
## CityPeterborough         -8.7099     1.5347  -5.675 1.49e-08 ***
## CitySault_Saint_Marie     0.7368     1.5347   0.480 0.631202
## CitySt_Catharines        -9.1967     1.5347  -5.993 2.26e-09 ***
## CitySudbury              -6.7386     1.5347  -4.391 1.16e-05 ***
## CityThunder_Bay          -1.5261     1.5347  -0.994 0.320089
## CityTimmins              -1.3246     1.5347  -0.863 0.388122
## CityToronto_East         -8.1949     1.5347  -5.340 9.85e-08 ***
## CityToronto_West         -7.3290     1.5347  -4.776 1.86e-06 ***
## CityWindsor              -9.6544     1.5347  -6.291 3.52e-10 ***
## poly(Time_Index, 3)1      4.3461    17.8974   0.243 0.808146
## poly(Time_Index, 3)2  -1096.4363    17.8974 -61.262  < 2e-16 ***
## poly(Time_Index, 3)3    798.5142    17.8974  44.616  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 17.9 on 3791 degrees of freedom
## Multiple R-squared:  0.6083, Adjusted R-squared:  0.6066
## F-statistic: 367.9 on 16 and 3791 DF,  p-value: < 2.2e-16
```

```
##
## Call:
## lm(formula = Price ~ City + poly(Time_Index, 3), data = auto_propane_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -21.1641  -5.4277   0.6259   4.3352  29.5266
##
```

```
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)            114.3491     0.4465 256.100   <2e-16 ***
## CityLondon             -14.2537     0.6309 -22.594   <2e-16 ***
## CityOttawa              13.0129     0.6309  20.627   <2e-16 ***
## CityThunder_Bay         -1.2574     0.6309  -1.993   0.0464 *
## CityToronto_East       -35.7780     0.6309 -56.712   <2e-16 ***
## CityToronto_West       -28.8232     0.6309 -45.688   <2e-16 ***
## CityWindsor            -22.9423     0.6309 -36.366   <2e-16 ***
## poly(Time_Index, 3)1  211.6931     7.3503  28.801   <2e-16 ***
## poly(Time_Index, 3)2 -181.3465     7.3503 -24.672   <2e-16 ***
## poly(Time_Index, 3)3  156.6610     7.3503  21.313   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.35 on 1893 degrees of freedom
## Multiple R-squared:  0.8556, Adjusted R-squared:  0.8549
## F-statistic:  1246 on 9 and 1893 DF,  p-value: < 2.2e-16
```

Section 8: Conclusion

Section 9: Appendix

Section 1: Introduction

Section 2: Data Description

Section 3: Data Preparation/Cleaning

```r
# Import needed libraries
library(tidyverse)
library(dplyr)
library(lubridate)
library(knitr)
library(ggplot2)
```

```r
# Get the data
raw_data = read.csv("data/fueltypesall.csv")
```

```r
# Rename the columns to full English
d = raw_data %>%
  rename(
    Toronto_West = Toronto.West.Ouest,
    Toronto_East = Toronto.East.Est,
    St_Catharines = St..Catharine.s,
    Sault_Saint_Marie = Sault.Saint.Marie,
    Thunder_Bay = Thunder.Bay,
    North_Bay = North.Bay,
    Parry_Sound = Parry.Sound,
    Ontario_Average = Ontario.Average.Moyenne.provinciale,
    Southern_Ontario_Average = Southern.Average.Moyenne.du.sud.de.l.Ontario,
    Northern_Ontario_Average = Northern.Average.Moyenne.du.nord.de.l.Ontario,
    Fuel_Type = Fuel.Type
  )
```

```r
# Show the column name changes in a comparison table
old_cols = names(raw_data)
new_cols = names(d)

names_comparison = data.frame(
  Before = old_cols,
  After = new_cols,
  stringsAsFactors = FALSE
)
kable(names_comparison, caption = "Old Column Names vs. New Column Names")
```

```r
# Drop the column Type.de.carburant
d = d %>%
  select(!Type.de.carburant)
```

```r
# Convert 0 values to NA
d = d %>%
  mutate(across(where(is.numeric), ~ na_if(., 0)))
```

```r
# Change the Date column type from char to date
d = d %>%
  mutate(Date = as.Date(Date))
```

```r
# Separate the data set into the six fuel categories and drop the Fuel_Type
# column for each new data set
regular_gas_data = d %>%
  filter(Fuel_Type == "Regular Unleaded Gasoline") %>%
  select(!Fuel_Type)

diesel_data = d %>%
  filter(Fuel_Type == "Diesel") %>%
  select(!Fuel_Type)

natural_gas_data = d %>%
  filter(Fuel_Type == "Compressed Natural Gas") %>%
  select(!Fuel_Type)

mid_gas_data = d %>%
  filter(Fuel_Type == "Mid-Grade Gasoline") %>%
  select(!Fuel_Type)

premium_gas_data = d %>%
  filter(Fuel_Type == "Premium Gasoline") %>%
  select(!Fuel_Type)

auto_propane_data = d %>%
  filter(Fuel_Type == "Auto Propane") %>%
  select(!Fuel_Type)
```

Section 4: Exploratory Data Analysis

```r
d %>%
  ggplot(aes(x = Date, y = Ontario_Average, color = Fuel_Type)) +
  geom_line() +
  labs(
    title = "Weekly Fuel Prices in Ontario by Fuel Type",
    x = "Week",
    y = "Average Fuel Price (cents/litre)",
    color = "Fuel Type")
```

Section 5: Statistical Inference

```r
### Filters the dataset for recent years (2021+) and runs paired t-tests to
### calculate the average regional price difference, confidence intervals, and
### statistical significance for gasoline, diesel, and auto propane.

# REGULAR GASOLINE
regional_compare_gas = regular_gas_data %>%
```

```r
  filter(Date >= as.Date("2021-01-01")) %>%
  select(Date, Northern_Ontario_Average, Southern_Ontario_Average) %>%
  drop_na() %>%
  mutate(Reg_Diff = Northern_Ontario_Average - Southern_Ontario_Average)

t_test_gas = t.test(regional_compare_gas$Northern_Ontario_Average,
                    regional_compare_gas$Southern_Ontario_Average,
                    paired = TRUE)
mean_diff_gas = t_test_gas$estimate
ci_lower_gas = t_test_gas$conf.int[1]
ci_upper_gas = t_test_gas$conf.int[2]
p_val_gas = t_test_gas$p.value

# DIESEL
regional_compare_diesel = diesel_data %>%
  filter(Date >= as.Date("2021-01-01")) %>%
  select(Date, Northern_Ontario_Average, Southern_Ontario_Average) %>%
  drop_na() %>%
  mutate(Reg_Diff = Northern_Ontario_Average - Southern_Ontario_Average)

t_test_diesel = t.test(regional_compare_diesel$Northern_Ontario_Average,
                       regional_compare_diesel$Southern_Ontario_Average,
                       paired = TRUE)
mean_diff_diesel = t_test_diesel$estimate
ci_lower_diesel = t_test_diesel$conf.int[1]
ci_upper_diesel = t_test_diesel$conf.int[2]
p_val_diesel = t_test_diesel$p.value

# AUTO PROPANE
regional_compare_propane = auto_propane_data %>%
  filter(Date >= as.Date("2021-01-01")) %>%
  select(Date, Northern_Ontario_Average, Southern_Ontario_Average) %>%
  drop_na() %>%
  mutate(Reg_Diff = Northern_Ontario_Average - Southern_Ontario_Average)

t_test_propane = t.test(regional_compare_propane$Northern_Ontario_Average,
                        regional_compare_propane$Southern_Ontario_Average,
                        paired = TRUE)
mean_diff_propane = t_test_propane$estimate
ci_lower_propane = t_test_propane$conf.int[1]
ci_upper_propane = t_test_propane$conf.int[2]
p_val_propane = t_test_propane$p.value
```

```r
### Compiles the calculated mean differences and 95% confidence intervals into a
### dataframe and generates a cleanly formatted summary table using the kable
## function.

# Create a dataframe to store the results
summary_table = data.frame(
  "Fuel Type" = c(
    "Regular Unleaded Gasoline",
    "Diesel",
    "Auto Propane"
```

```r
  ),
  "Mean Difference (cents/litre)" = c(
    round(mean_diff_gas, 2),
    round(mean_diff_diesel, 2),
    round(mean_diff_propane, 2)
  ),
  "95% Confidence Interval" = c(
    paste0("(", round(ci_lower_gas, 2), ", ", round(ci_upper_gas, 2), ")"),
    paste0("(", round(ci_lower_diesel, 2), ", ", round(ci_upper_diesel, 2),
           ")"),
    paste0("(", round(ci_lower_propane, 2), ", ", round(ci_upper_propane, 2),
           ")")
  ),
  check.names = FALSE # This prevents R from putting periods in the column names
)

# Use kable to print a perfectly formatted table
kable(summary_table,
      caption = "Average Price Differences: Northern vs. Southern Ontario",
      align = c("c", "c", "c"))
```

```r
### Compiles the p-values from the paired t-tests into a summary table to
### evaluate the statistical significance of regional price differences across
### the three fuel types.

# Dataframe for the Hypothesis Testing results
ht_table = data.frame(
  "Fuel Type" = c(
    "Regular Unleaded Gasoline",
    "Diesel",
    "Auto Propane"
  ),
  "p-value" = c(
    format.pval(p_val_gas, digits = 3, eps = 0.001),
    format.pval(p_val_diesel, digits = 3, eps = 0.001),
    format.pval(p_val_propane, digits = 3, eps = 0.001)
  ),
  check.names = FALSE
)

# Table
kable(ht_table,
      caption = "Paired T-Test Results: Regional Price Differences",
      align = c("l", "c", "c"))
```

Section 6: Bootstrapping

```r
### Mimics the concept of repeated sampling by using the bootstrap method to
### generate distributions and 95% confidence intervals for regional
### price differences across the three fuel types.
```

```r
set.seed(57)
B = 1000      # Number of bootstrap iterations

get_boot_ci = function(df) {

  boot_function = function() {
    boot_data = df %>% slice_sample(n = nrow(df), replace = TRUE)
    boot_mean = boot_data %>% summarize(cal.s = mean(Reg_Diff, na.rm = TRUE))
    return(boot_mean[1,1])
  }

  # Replicate the process B times
  boot_means = replicate(B, boot_function())

  # 2.5% and 97.5% percentiles for the 95% CI
  ci = quantile(boot_means, probs = c(0.025, 0.975))
  return(c(lower = ci[1], upper = ci[2]))
}

# Apply the function to all three fuel types using the dataframes we made in
# Section 5
boot_gas = get_boot_ci(regional_compare_gas)
boot_diesel = get_boot_ci(regional_compare_diesel)
boot_propane = get_boot_ci(regional_compare_propane)
```

```r
### Compiles the 95% confidence intervals from both the classical t-tests and
### the bootstrap resampling into a comparative dataframe to demonstrate the
### stability of the regional price difference estimates across all three fuel
### types.

boot_table = data.frame(
  "Fuel Type" = c("Regular Unleaded Gasoline", "Diesel", "Auto Propane"),
  "Classical 95% CI (T-Test)" = c(
    paste0("(", round(ci_lower_gas, 2), ", ", round(ci_upper_gas, 2), ")"),
    paste0("(", round(ci_lower_diesel, 2), ", ",
           round(ci_upper_diesel, 2), ")"),
    paste0("(", round(ci_lower_propane, 2), ", ",
           round(ci_upper_propane, 2), ")")
  ),
  "Bootstrap 95% CI" = c(
    paste0("(", round(boot_gas["lower.2.5%"], 2), ", ",
           round(boot_gas["upper.97.5%"], 2), ")"),
    paste0("(", round(boot_diesel["lower.2.5%"], 2), ", ",
           round(boot_diesel["upper.97.5%"], 2), ")"),
    paste0("(", round(boot_propane["lower.2.5%"], 2), ", ",
           round(boot_propane["upper.97.5%"], 2), ")")
  ),
  check.names = FALSE
)

kable(boot_table, caption =
        "Comparison of Classical and Bootstrapped 95% Confidence Intervals",
      align = c("l", "c", "c"))
```

```r
#Creates a function to extract the 1000 raw means for plotting
get_boot_means = function(df) {
  boot_func = function() {
    boot_data = df %>% slice_sample(n = nrow(df), replace = TRUE)
    boot_mean = boot_data %>% summarize(cal.s = mean(Reg_Diff, na.rm = TRUE))
    return(boot_mean[1,1])
  }
  return(replicate(B, boot_func()))
}

# Generates the 1,000 means for the three fuel types
boot_means_gas = get_boot_means(regional_compare_gas)
boot_means_diesel = get_boot_means(regional_compare_diesel)
boot_means_propane = get_boot_means(regional_compare_propane)

#Combines everything into one large dataframe AND set specific order (Levels)
plot_data = data.frame(
  Fuel_Type = factor(
    rep(c("Regular Unleaded Gasoline", "Diesel", "Auto Propane"), each = B),
    levels = c("Regular Unleaded Gasoline", "Diesel", "Auto Propane")
  ),
  Means = c(boot_means_gas, boot_means_diesel, boot_means_propane)
)

#Creates a dataframe for the Confidence Intervals with the same ordered factor
ci_data = data.frame(
  Fuel_Type = factor(
    c("Regular Unleaded Gasoline", "Diesel", "Auto Propane"),
    levels = c("Regular Unleaded Gasoline", "Diesel", "Auto Propane")
  ),
  Lower = c(boot_gas[1], boot_diesel[1], boot_propane[1]),
  Upper = c(boot_gas[2], boot_diesel[2], boot_propane[2])
)

#Plot using facet_wrap, adding colors and forcing 3 columns (left-to-right)
ggplot(plot_data, aes(x = Means, fill = Fuel_Type)) +
  geom_histogram(bins = 40, color = "black", linewidth = 0.2, alpha = 0.8,
                 show.legend = FALSE) +
  geom_vline(data = ci_data, aes(xintercept = Lower), color = "red",
             linetype = "dashed", linewidth = .4) +
  geom_vline(data = ci_data, aes(xintercept = Upper), color = "red",
             linetype = "dashed", linewidth = .4) +
  scale_fill_manual(values = c(
    "Regular Unleaded Gasoline" = "steelblue",
    "Diesel" = "darkorange",
    "Auto Propane" = "forestgreen"
  )) +
  facet_wrap(~ Fuel_Type, scales = "free_x", ncol = 3) +
  labs(
    title = "Bootstrap Distributions of Mean Regional Differences",
    subtitle = "Northern Avg - Southern Avg (1,000 Resamples)",
    x = "Mean Difference (cents/litre)",
    y = "Frequency"
```

```r
  ) +
  theme_minimal(base_size = 14, base_family = "serif") +
  theme(strip.text = element_text(face = "bold", size = 10))
```

Section 7: Regression Analysis

```r
# Data split
regular = regular_gas_data %>%
  mutate(group_ind = sample(c(1:5), size = nrow(regular_gas_data), replace = TRUE))
diesel = diesel_data %>%
  mutate(group_ind = sample(c(1:5), size = nrow(regular_gas_data), replace = TRUE))
propane = auto_propane_data %>%
  mutate(group_ind = sample(c(1:5), size = nrow(auto_propane_data), replace = TRUE))


regular_mse = vector()
diesel_mse = vector()
propane_mse = vector()

for(d in 1:5) {
  mse_fold = vector()

  for(i in 1:5) {

    train_set = regular %>% filter(group_ind != i)
    validation_set = regular %>% filter(group_ind == i)

    m_train = lm(Price ~ City + poly(Time_Index, d), data = train_set)
    y_hat = predict(m_train, newdata = validation_set)
    mse_fold[i] = mean((validation_set$Price - y_hat)^2, na.rm = TRUE)
  }

  # Take average as final score
  regular_mse[d] = mean(mse_fold)
}


cv_result_regular = data.frame(
  "Polynomial Degree" = c("Degree 1", "Degree 2", "Degree 3", "Degree 4", "Degree 5"),
  "Cross-Validated MSE" = regular_mse,
  check.names = FALSE
)

for(d in 1:5) {
  mse_fold = vector()

  for(i in 1:5) {

    train_set = diesel %>% filter(group_ind != i)
    validation_set = diesel %>% filter(group_ind == i)

    m_train = lm(Price ~ City + poly(Time_Index, d), data = train_set)
```

```r
    y_hat = predict(m_train, newdata = validation_set)
    mse_fold[i] = mean((validation_set$Price - y_hat)^2, na.rm = TRUE)
  }

  diesel_mse[d] = mean(mse_fold)
}

cv_result_diesel = data.frame(
  "Polynomial Degree" = c("Degree 1", "Degree 2", "Degree 3", "Degree 4", "Degree 5"),
  "Cross-Validated MSE" = diesel_mse,
  check.names = FALSE
)

for(d in 1:5) {
  mse_fold = vector()

  for(i in 1:5) {
    # Explicitly defining Train and Validation sets per fold
    train_set = propane %>% filter(group_ind != i)
    validation_set = propane %>% filter(group_ind == i)

    m_train = lm(Price ~ City + poly(Time_Index, d), data = train_set)
    y_hat = predict(m_train, newdata = validation_set)
    mse_fold[i] = mean((validation_set$Price - y_hat)^2, na.rm = TRUE)
  }

  propane_mse[d] = mean(mse_fold)
}

cv_result_propane = data.frame(
  "Polynomial Degree" = c("Degree 1", "Degree 2", "Degree 3", "Degree 4", "Degree 5"),
  "Cross-Validated MSE" = propane_mse,
  check.names = FALSE
)



kable(cv_result_regular, caption = "Cross-Validation Error (Regular Gas)")
kable(cv_result_diesel, caption = "Cross-Validation Error (Diesel)")
kable(cv_result_propane, caption = "Cross-Validation Error (Auto Propane)")

plot_cv_data = data.frame(
  Degree = rep(1:5, 3),
  MSE = c(regular_mse, diesel_mse, propane_mse),
  Fuel_Type = rep(c("Regular Gas", "Diesel", "Auto Propane"), each = 5)
)

ggplot(plot_cv_data, aes(x = Degree, y = MSE, color = Fuel_Type, group = Fuel_Type)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  scale_color_manual(values = c("Regular Gas" = "blue",
                                "Diesel" = "red",
                                "Auto Propane" = "orange")) +
```

```r
  labs(
    title = "Cross-Validation Error by Polynomial Degree",
    x = "Polynomial Degree",
    y = "MSE",
    color = "Fuel Type"
  )
```

```r
# Final models
regular_fit = lm(Price ~ City + poly(Time_Index, 3), data = regular_gas_data)
diesel_fit = lm(Price ~ City + poly(Time_Index, 3), data = diesel_data)
propane_fit = lm(Price ~ City + poly(Time_Index, 3), data = auto_propane_data)
summary(regular_fit)
summary(diesel_fit)
summary(propane_fit)
```

Section 8: Conclusion